PADHAKU
Curiosity · Class 6 Science



CHAPTER 11

Nature's Treasures

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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⭐ Introduction

? Ajji asks: can you think of more such treasures of nature?

💡 SAMPLE ANSWER

Air, water, sunlight, soil, forests (trees and plants), animals, rocks, minerals and fossil fuels like coal and petroleum.

⭐ 11.1 Air

Activity 11.1 **Breathing exercise: take a deep breath, hold it as long as you can and breathe out slowly. How long can you hold your breath? How do you feel when you hold your breath?**

💡 SAMPLE ANSWER

Observation: Most of us can hold our breath only for a short time (about 20–40 seconds). We feel uneasy and want to breathe again.

Conclusion: The air we breathe in has oxygen. Our body needs oxygen all the time to work, so we cannot hold our breath for long.

(Caution: do not hold your breath until you feel uncomfortable.)

? **Can you name some gases which are present in the air?**

✔ ANSWER

Nitrogen (78%), oxygen (21%), and argon, carbon dioxide and other gases (1%).

Activity 11.2 **Make a firki (paper pinwheel). Hold it and run, or blow air on it. What do you observe? Does the firki rotate? What makes a firki rotate?**

✔ ANSWER

Observation: Yes, the firki rotates when we run with it, blow on it or move it back and forth.

Conclusion: Moving air (wind) makes the firki rotate. A windmill works in the same way.

? **Find out more other windmill farms in our country.**

💡 SAMPLE ANSWER

Besides Muppandal (Tamil Nadu), Jaisalmer (Rajasthan) and Brahmanvel (Maharashtra), there are wind farms in Gujarat (e.g. Kutch), Karnataka and Andhra Pradesh.

⭐ 11.2 Water

? How do you feel when you cannot get water to drink, especially when you are thirsty?

💡 SAMPLE ANSWER

We feel uneasy, tired and weak; the mouth and throat feel dry. Water is precious for us.

? Can you think of some more uses of water in your daily life? (We need water for drinking...)

✔ ANSWER

We need water for cooking, bathing, washing clothes and utensils, cleaning the house, watering plants and giving water to animals.

? Where do we get water from? Make a list of the different sources of water.

✔ ANSWER

- Rain
- Rivers and streams
- Lakes and ponds
- Wells, tube wells and hand pumps (underground water)
- Ice and snow on mountains
- Oceans and seas (salty water)

? **Do you feel that water is being used efficiently in our daily activities? Have you observed water being wasted in your daily activities?**

💡 **SAMPLE ANSWER**

No, not always. Water is often wasted, e.g. taps left running while brushing or washing, leaking taps and pipes, and overflowing tanks.

Activity 11.3 **Fill Column II (How is water wasted?) and Column III (Suggest ways to reduce wastage of water) in Table 11.1.**

💡 **SAMPLE ANSWER**

Activity	How is water wasted?	Ways to reduce wastage
1. Hand washing	Tap kept running while rubbing soap	Close the tap while applying soap
2. Washing clothes	Clothes rinsed under running water; machine run half empty	Use buckets; run the machine with a full load; reuse rinse water for cleaning floors
3. Washing utensils	Tap left open throughout	Wash in a tub or bucket; close the tap while scrubbing
4. Taking shower	Long showers	Use a bucket and mug; take short showers

5. Cooking	Vegetables and rice washed under running water	Wash in a bowl; use that water for plants
6. Gardening	Watering with a pipe in the hot afternoon; water flows away	Water in the morning or evening; use a can or drip irrigation
7. Brushing teeth	Tap left running	Use a mug of water; close the tap

Conclusion: A lot of water is wasted in daily work. Closing taps, fixing leaks and reusing water can save a lot of it.

? **What conclusion can you draw from the table? What can you and your family do to reduce this wastage of water?**

💡 **SAMPLE ANSWER**

We waste a lot of water carelessly every day. We can turn off taps when not in use, fix leaking taps and pipes, use buckets instead of showers or pipes, reuse water (e.g. water used for washing vegetables for plants) and harvest rainwater.

? **Identify other human activities that lead to water pollution. Discuss what you can do to reduce water pollution.**

💡 **SAMPLE ANSWER**

Activities: throwing garbage, plastic and flowers into rivers and ponds; washing clothes, vehicles and animals in rivers; letting sewage and factory waste into rivers; too many fertilisers and pesticides from farms washing into water.

We can: not throw waste into water bodies, use dustbins, use fewer chemicals and spread awareness in our area.

? In what ways can you conserve water?

✔ ANSWER

- Turn off taps when not in use
- Fix leaking taps and pipes
- Reuse water (e.g. for plants or cleaning)
- Harvest rainwater
- Do not pollute water sources

? Find traditional water harvesting practices in your locality. Discuss with your teachers and parents.

💡 SAMPLE ANSWER

Examples in India: stepwells (Bawadi in Rajasthan, Vav in Gujarat), johads and kunds (Rajasthan), tanks and ponds (eri in Tamil Nadu), bamboo drip irrigation (Meghalaya). Write the one used in your area.

Do you know?

World Water Day is observed on 22nd March every year. Find out its importance.

💡 SAMPLE ANSWER

It reminds people that freshwater is precious and limited. It spreads awareness about saving water, keeping it clean and providing safe drinking water to everyone.

⭐ 11.3 Energy from the Sun

?

Bhoomi draws pictures to show uses of heat and light from the Sun (Plants make food, Drying of clothes). Help her by adding more examples.

💡 SAMPLE ANSWER

- Drying chillies, grains and papad
- Making pickles and drying raw mangoes
- Making salt from sea water
- Cooking in a solar cooker and heating water in a solar water heater
- Producing electricity using solar panels
- Light during the day so we can see and work
- Keeping us warm in winter

? **Read the conversation of Bhoomi and Surya about the cow grazing grass. According to you, whose statement is correct and why?**

✔ ANSWER

The last statement is correct: **the cow gets energy from the Sun indirectly.** The cow gets energy by eating grass, and the grass makes its food using sunlight. So the Sun is the main source of energy. Standing in the Sun does not give the cow energy.

? **What will happen if the Sun is not visible for a few days? (1. We may have to depend on artificial lighting during day time also. 2. ... 3. ...)**

💡 SAMPLE ANSWER

2. Wet clothes, grains and other things will not dry easily.

3. Plants will not be able to make enough food, and it will become colder. Solar panels and solar cookers will not work.

⭐ 11.4 Forests

? **Discuss with your friends and make a list of at least five products that we get from forests.**

✔ ANSWER

- Wood (timber) and firewood
- Fruits, nuts and honey
- Medicinal plants (herbs)
- Gum, resin and lac
- Bamboo, leaves and fodder for animals
- Paper (made from wood)

? **What are the consequences of cutting a large forest area? Make a presentation, role play, story or poem showing what could happen if we continue to cut down trees in our forests.**

💡 **SAMPLE ANSWER**

Consequences:

- Animals and birds lose their home and food; many may die out
- Soil gets washed away by rain and loses fertility
- Less rainfall, more floods and droughts
- More carbon dioxide and less fresh air; the place becomes hotter
- People lose products like wood, fruits and medicines

Short poem: “Cut the trees, the birds will cry, / The soil will wash, the land go dry; / Plant a tree and let it grow, / Fresh air and rain will surely flow.”

? **You too can plan a Van Mahotsav in your community.**

💡 **SAMPLE ANSWER**

Plan: choose a date in July, get permission and saplings (from a nursery or forest department), pick safe places, dig pits, plant saplings with friends and elders, water them, put guards around them and give each sapling a caretaker.

⭐ 11.5 Soil, Rocks and Minerals

? **Do you realise that earthworms are natural agents that help in turning and loosening the soil?**

✔ **ANSWER**

Yes. Earthworms make tunnels in the soil, which loosens it so that air and water can reach the roots. Their waste also makes the soil fertile.

Activity 11.4 **Collect soil samples from different places. Guess what could be in them, observe colour and texture with naked eye and a magnifying lens, and record in Table 11.2.**

💡 **SAMPLE ANSWER**

Location	What I guessed	Naked eye (colour, texture)	With magnifying lens
Soil from a farm after ploughing	Soil, small stones	Dark brown, soft, a bit moist	Tiny particles, roots, small insects
Garden	Soil, leaves	Dark brown/black, soft	Bits of dead leaves, small worms
Roadside	Dust, stones	Light brown, dry, rough	Sand grains, small pebbles, bits of plastic
Riverbank or pond	Mud	Grey, sticky, wet	Very fine particles, small shells

? **Is there any difference between your guess and what you can actually observe when you look closely?**

💡 **SAMPLE ANSWER**

Yes. When we look closely we see more things than we guessed, like tiny insects, roots, bits of leaves and small pebbles.

? **Do you see any differences in the soil samples taken from different places?**

✔ **ANSWER**

Yes. They differ in colour (black, brown, red, grey), texture (soft, sticky, sandy) and in the things they contain, because they contain different materials.

? **Do you see differences in what you are able to observe with your naked eye and what you can observe with a magnifying lens?**

✔ ANSWER

Yes. With a magnifying lens we can see small particles, tiny insects and plant bits that we cannot see with the naked eye. Some organisms are so small that we cannot see them even then.

⭐ 11.6 Fossil Fuels

? **Surya, why do different types of vehicles go to different filling stations?**

✔ ANSWER

Different vehicles use different types of fuels, for example petrol, diesel or CNG, so they go to stations that sell their fuel.

Activity 11.5 **Conduct a survey of vehicles in your neighbourhood. Which types of vehicles are there? What types of fuels do they use? Record in Table 11.3.**

💡 SAMPLE ANSWER

Type of vehicle	Type of fuel used
Scooter, motorcycle	Petrol
Car	Petrol, diesel or CNG
Bus	Diesel or CNG
Truck, tractor	Diesel
Auto-rickshaw	CNG or LPG
E-rickshaw, electric scooter	Electricity (battery)

Conclusion: Petrol and diesel are the most commonly used fuels.

? **What are the most common types of fuels used?**

✔ ANSWER

Petrol and diesel.

? **Find out the major coal-producing states and mark them in a map of India.**

💡 SAMPLE ANSWER

Major coal-producing states: Jharkhand, Odisha, Chhattisgarh, Madhya Pradesh, West Bengal, Telangana and Maharashtra. Mark them on an outline map of India.

? Let us do our bit to conserve fossil fuels by walking or cycling to nearby places and using public transport. Suggest some more ways.

💡 SAMPLE ANSWER

- Share a vehicle (car pool) with others
- Switch off the vehicle engine at red lights
- Keep tyres properly inflated and vehicles well serviced
- Use electric vehicles and solar energy
- Save electricity, since much of it is made from coal

⭐ 11.8 Resources we use

? Can you list some alternatives for reducing air pollution?

✔ ANSWER

- Use electric vehicles, CNG and public transport
- Walk or cycle for short distances
- Use solar and wind energy instead of fossil fuels
- Do not burn garbage or crop waste

- Plant more trees

Activity 11.6 **Make a list of activities you do in your daily life and write the natural resources used directly or indirectly (Table 11.4).**

💡 **SAMPLE ANSWER**

Activity	Natural resource
Washing clothes	Water
Making clay toys	Soil (clay), water
Collecting firewood	Forest (trees)
Making kites	Bamboo (forest), paper (from wood), air (to fly)
Having breakfast	Food from plants and animals, water
Cooking food	LPG/natural gas, water
Drying clothes	Sunlight, air

? **How many natural resources did you list? Compare your list with that of your friend.**

💡 **SAMPLE ANSWER**

In the table above about 8 natural resources are listed: water, soil, forest (trees), air, sunlight, plants, animals and natural gas. Your friend's list may have some different ones.

Let us enhance our learning

1 **Fig. 11.9 shows items related to natural resources. Match them with their jumbled up names. Make another table and write the names of these resources. Classify them as renewable or non-renewable.**

Jumbled names: ocrk, refost, ndiw, atwre

✔ ANSWER

Jumbled name	Resource	Type
ocrk	Rock	Non-renewable
refost	Forest	Renewable
ndiw	Wind (air)	Renewable
atwre	Water	Renewable

2 (i) **Nature has all the resources to meet human needs. [T/F]**

✔ ANSWER

T (True)

2 (ii) **Machines are a resource found in nature. [T/F]**

✔ ANSWER

F (False). Correct: Machines are human-made resources.

2 (iii) **Natural gas is a non-renewable resource. [T/F]**

✔ ANSWER

T (True)

2 (iv) **Air is a renewable resource. [T/F]**

✔ ANSWER

T (True)

3 (i) **A fuel that is commonly used in two wheelers like scooters or bikes is ____. (a) Kerosene (b) Petrol (c) Diesel (d) LPG**

✔ ANSWER

(b) Petrol. Scooters and motorcycles run on petrol.

3 (ii) **An example of a renewable resource is ____. (a) Coal (b) Water (c) Natural gas (d) Petrol**

✔ ANSWER

(b) Water. Nature renews water through rain; the others are fossil fuels.

4 **Classify the following as renewable or non-renewable resources—coal, natural gas, forests and minerals.**

✔ ANSWER

Renewable: forests

Non-renewable: coal, natural gas, minerals

5 **Why do we say that petroleum is a non-renewable resource?**

✔ ANSWER

Petroleum takes millions of years to form from remains of microorganisms and plants buried deep in the earth. It is found in limited quantity and, once used, it is not replenished within a reasonable time. So it is non-renewable.

6 **It is difficult to regrow forests. Justify this statement.**

✔ ANSWER

Trees grow slowly; it takes many years for a sapling to become a big tree. A forest also has many kinds of plants, animals and rich soil that take a long time to develop. So once a forest is cut, it takes many years to grow a new forest or restore it.

7 **Make a list of five daily activities in which you use natural resources. Suggest ways by which you can reduce their use.**

💡 **SAMPLE ANSWER**

Activity	Resource	How to reduce use
Brushing teeth	Water	Use a mug; close the tap
Bathing	Water	Use a bucket, not a shower
Cooking	LPG (natural gas)	Cover pots; use a pressure cooker
Going to school	Petrol/diesel	Walk, cycle or use the school bus
Studying / writing	Paper (trees), electricity	Use both sides of paper; switch off lights and fans when not needed

8 **List four activities that are possible due to the presence of air.**

✔ **ANSWER**

- Breathing (we get oxygen)

- Flying kites and rotating a firki
- Drying of wet clothes
- Running windmills to make electricity (also burning of fuel needs air)

9 **How can you contribute towards enhancing the green cover of your locality? Make a list of actions to be taken.**

💡 **SAMPLE ANSWER**

- Plant saplings on birthdays and during Van Mahotsav
- Water and protect young plants with tree guards
- Do not cut or harm trees; stop others from doing so
- Grow plants in pots, balconies and school gardens
- Save paper so fewer trees are cut
- Spread awareness through an eco-club

10 (i) **In the given illustration, food is being cooked. What type of energy is being used for cooking? (check the picture in your book)**

💡 **SAMPLE ANSWER**

Solar energy (energy from the Sun) is used, in a solar cooker.

10 (ii) **Name one benefit and one drawback of using this type of energy for cooking.**

✔ ANSWER

Benefit: It is free, does not cause pollution and saves fuel.

Drawback: It cannot be used at night or on cloudy days, and cooking takes more time.

11 **Cutting down trees on a large scale impacts the quality of the soil. Why do you think it is so?**

✔ ANSWER

Roots of trees hold the soil. Without trees, rain and wind wash away the top fertile soil. Also, no fallen leaves decay to add nutrients to the soil. So the soil becomes poor (less fertile).

12 **Explain two ways in which human activities pollute the air. Propose one action which can help in reducing air pollution.**

✔ ANSWER

Ways:

1. Burning fossil fuels like petrol and diesel in vehicles releases smoke and carbon dioxide.

2. Factories and burning of garbage or crop waste release smoke and harmful gases.

Action: Use public transport, bicycles or electric vehicles instead of private petrol/diesel vehicles.

13 **A family uses solar panels to generate electricity, a gas stove to cook food and a windmill for pumping water from a well. What would happen if there were no sunlight for a week?**

💡 SAMPLE ANSWER

- The solar panels would not make electricity, so the family would have no (or little) electricity.
- The gas stove would still work, so they could cook food.
- The windmill could still pump water if wind blows. (Over a long time, less sunlight also means less wind and plants cannot make food.)

14 **Fill up the blanks using the following terms— (fossil fuels, forest, air, petroleum, coal, water and non-renewable resource)**

Concept map: Natural Resources → Renewable Resources → ___, ___, ___; Natural Resources → ___ → ___ → ___, ___, Natural gas

💡 SAMPLE ANSWER

- **Natural Resources → Renewable Resources → Air, Water, Forest**

- **Natural Resources → Non-renewable resource → Fossil fuels → Petroleum, Coal**, Natural gas

(check the picture in your book for the exact boxes)

15 **There is an increasing demand of trees to meet the requirements of industries and for housing. Therefore, trees are being felled. Is it justified? Discuss and prepare a brief report.**

💡 SAMPLE ANSWER

Report: Cutting trees for industries and housing

No, cutting trees on a large scale is not justified. We need wood and land, but forests give us fresh air, rain, fertile soil and homes for animals. Cutting them causes soil erosion, floods, heat and air pollution, and a forest takes many years to regrow.

Suggestions: cut only as many trees as needed; plant more trees than are cut; use alternatives like bamboo, recycled wood and steel; recycle paper; build on already-cleared land.

16 **Propose a plan to use less water in your school. What steps would you take to make this plan happen and how would it help the environment?**

💡 SAMPLE ANSWER

Plan:

- Check and repair leaking taps and pipes
- Put “Close the tap” posters near taps
- Use mugs/bottles for drinking instead of open taps
- Use water left in bottles and from washing hands for plants
- Set up rainwater harvesting on the school roof
- Form a water monitor team in each class

Help to environment: saves freshwater, keeps more groundwater, reduces electricity used for pumping water, and leaves enough water for everyone.

📘 Learning further

1 **Find out some of the traditional rainwater harvesting techniques being used in your state or in other parts of the country.**

💡 SAMPLE ANSWER

- Stepwells: Bawadi (Rajasthan), Vav (Gujarat)
- Johads, kunds and taankas (Rajasthan)
- Eris (tanks) in Tamil Nadu

- Zabo system (Nagaland)
- Bamboo drip irrigation (Meghalaya)
- Surangam (Kerala, Karnataka)

2 **Investigate the effect of air pollution on human health by talking to elders. Identify the main sources of air pollution in your area and suggest two practical steps your school or community could take.**

💡 SAMPLE ANSWER

Effects on health: cough, breathing problems like asthma, burning eyes, headache, lung and heart diseases.

Sources: vehicles, factories, burning garbage and crop waste, dust from construction.

Two steps: (1) a 'no garbage burning' awareness drive; (2) plant trees and encourage walking, cycling or car pooling to school.

3 **Prepare a list of the names and uses of important minerals and rocks used in your village/town/city.**

💡 SAMPLE ANSWER

Mineral / rock	Use
Granite	Floors, kitchen tops, buildings
Marble	Floors, statues, temples
Sandstone	Buildings, walls
Slate	Roofing, writing slates

Laterite	Building blocks like bricks
Iron	Tools, utensils, buildings
Copper	Electric wires, utensils
Gold, silver	Jewellery, mobile phones

4 **You are an eco-club monitor. Organise a tree plantation drive in your school. List the steps and prepare a one-page report listing trees planted with their importance.**

💡 SAMPLE ANSWER

Steps: take the teacher's and principal's permission → choose the place and date → get saplings and tools → dig pits → plant and water → put tree guards → give each sapling a caretaker.

Tree planted	Importance
Neem	Medicinal, gives shade, cleans air
Peepal	Shade, home for birds
Amla (nellikai)	Fruits rich in vitamin C, medicinal
Mango	Fruits, shade
Banyan	Big shade, home for birds and animals

🎉 **Chapter 11 done! Revise the answers once more tomorrow.**